# Project Report


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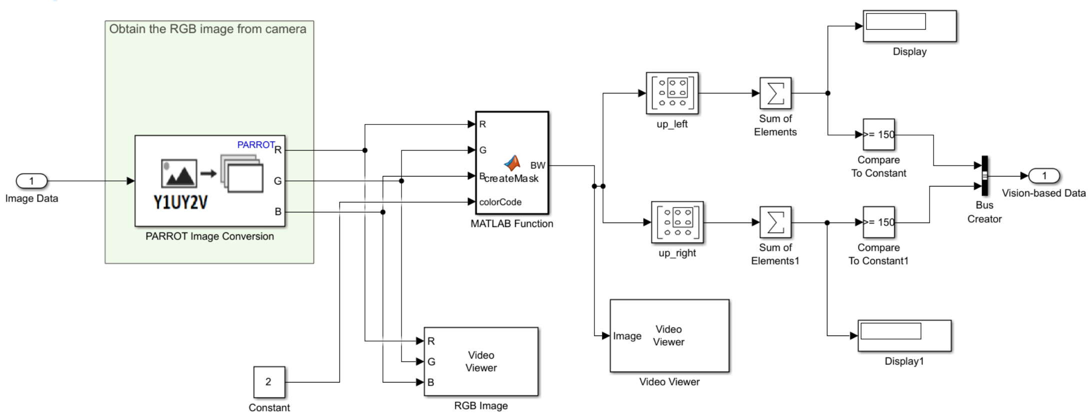
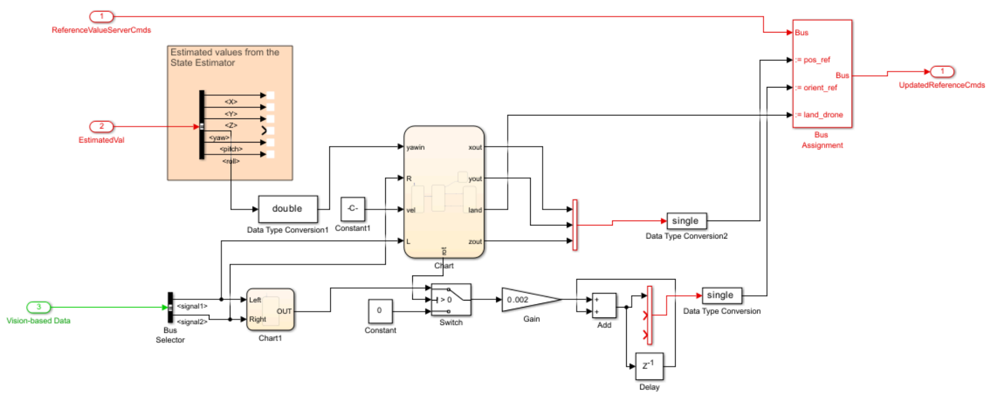
## 1. Simulink Model

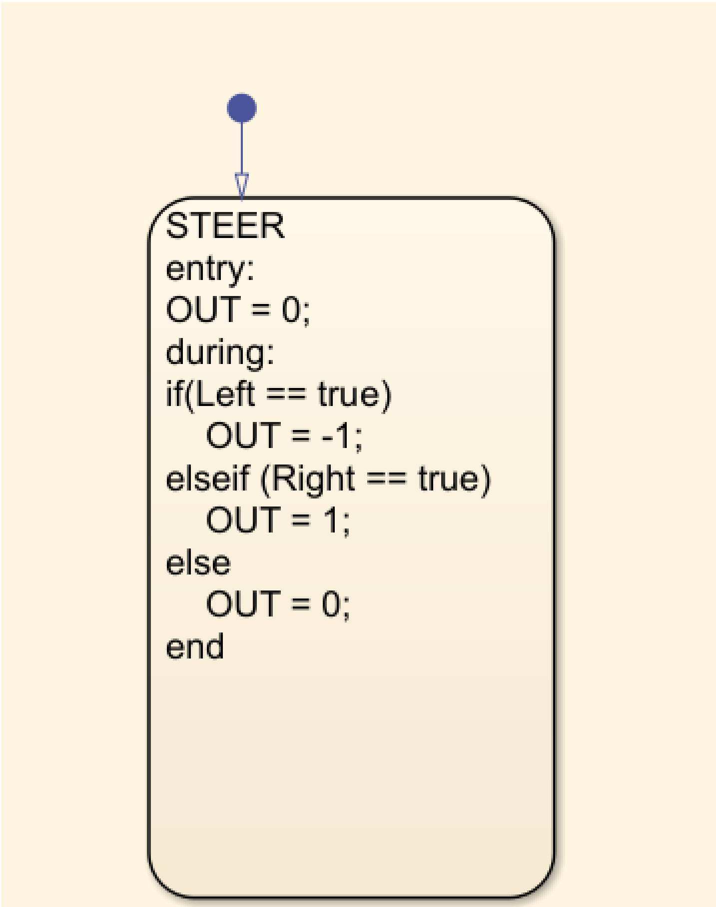
- To make the simulink model for this lab, I modified the model I have used in lab 3 for platform following purposes.
- In the Image Processing System of the Simulink model in the project, I used the same color reading model from the last lab to read the block through the camera. However, I modified it so that it differentiates between whether it detected the color of the platform or the floor. This will keep the drone to constantly turn left and right to the color platform in order to keep it on the line while moving forward.



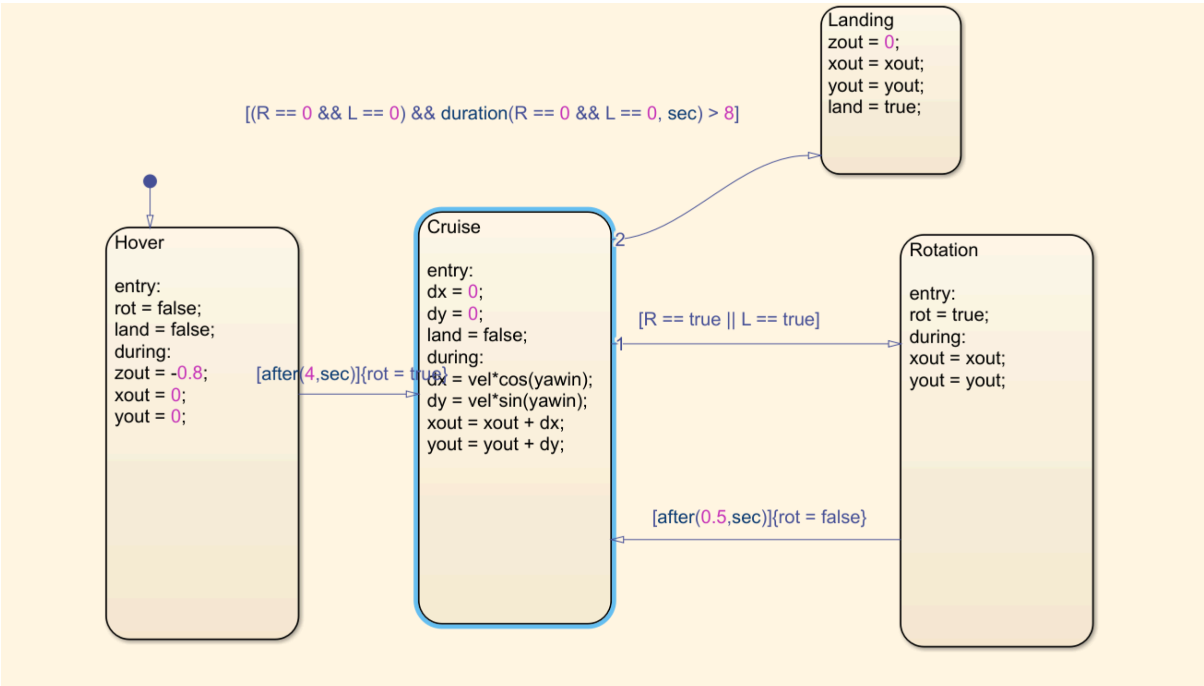

- In the path planning of the control system, I will modify it to make it look like the model below.




- The chart1 block is for steering to make the drone steer to follow the platform




- The Chart block is for motion control, which will be decided by the left and right signal from the color detection. For the landing, if it detects the color for more than 8 seconds, it will land.




## 2. Line following cart

- I am in a group of 3 sharing a line-following cart. It follows nicely for a while but gets very easy to drift away.

### 3. Flight Control

- I put the drone on the start of the line and started the Simulink. The drone took off, hovered while detecting the color platform before starting to move while following it. When it reached the end, it landed down.